# Bill of Materials

| Item | Model | Description | Make | Qty |
|---|---|---|---|---|
| 1 | Item 1 | Desc 1 | Make 1 | 1 |
| 2 | Item 2 | Desc 2 | Make 2 | 2 |

# Item 1: Desc 1

**Make: Make 1 | Model: Item 1 | Qty: 1**

## Datasheet

# MUHD IKMAL HAKIMI BIN MUHAMMAD RASLI

**BEng Mechanical Engineering**

7, Topgreen Scenery Villa, Jln Stutong, 93350 Kuching, Sarawak

ikmalrasli@gmail.com • 010 905 7747


## EDUCATION

Sep 2019 – Present
**BEng Mechanical Engineering, Universiti Malaya**
Final CGPA: 3.63 MUET: Band 4

July 2018 – April 2019
**Foundation in Physical Science, Centre for Foundation Studies Universiti Malaya**
Final CGPA: 4.00

Feb 2016 – Nov 2017
**Sekolah Menengah Kebangsaan Batu Lintang**
SPM: 5A+, 2A, 2A-, 1B+

## WORK EXPERIENCE

Jan – Feb 2024
**Substitute school Teacher**
Performed all necessary duties of a school teacher, carried out any ad-hoc work given by the school.

- **Skills learned –** teamwork, attention to detail, problem-solving skills, communication skills, time management.

Aug – Oct 2022
**Industrial training**

- Training done under Sarawak Energy Berhad (SEB), placed at Sejingkat Power Corporation (SPC), a 210MW coal fired power plant.
- Worked on weekly equipment status checks, observed maintenance duties, and learned of overall process of the powerplant system.
- **Skills learned** – multi-tasking, teamwork, problem-solving skills, communication skills, time management.

2019-2020
**Emcee**

- Worked on emcee duties during Frenster (College event).
- **Skills learned** – communication skills, attention to detail.

2018-2020
**Illustrator**

- Worked on posters for activities during foundation and first year such as KEMAS and Frenster (College events).
- **Skills learned** – communication skills, attention to detail.

## PROJECT EXPERIENCE

2023
**Development of Automatic Fish Feeder for Aquaponic Systems**

- Developed an automatic fish feeder that dispenses appropriate amounts of food based on fish number and age.

2022
**RFID Home Security System with License Plate Recognition**

- Automatic gate that opens by detecting RFID and license plate of the house owner vehicle through image processing and machine learning process.

2021
**Fire Alarm System**

- A system that detects fire through temperature, light and gas sensors, Arduino as microcontroller, and LabVIEW.

2020
**Programming Workshop**

- Provided a two week workshop that tutors secondary school students programming, in C++.

## ADDITIONAL SKILLS

- Mandarin language (6 years formal study)
- Japanese language (2 years self study)
- Programming (Python, JAVA, C++)
- Web development (HTML, CSS, Javascript, Angular)
- ANSYS Fluent
- Solidworks
- MATLAB

## REFERENCE

**Dr. Ahmad Saifizul bin Abdullah**
Department of Mechanical Engineering,
Faculty of Engineering,
Universiti Malaya

+603-79674597
saifizul@um.edu.my

**Shahrizal bin Mustapha**
IMS Specialist,
IMSA Division,
CSS Department,
Sarawak Energy Berhad

+6012-7270424
sharizal.mustapha@sarawakenergy.com

# Item 2: Desc 2

**Make: Make 2 | Model: Item 2 | Qty: 2**

## Datasheet

# Introduction:

UM graduate (3.63 CGPA) → Bachelor's degree in Mechanical Engineering

↓

Have experience with programming → Web development (HTML, CSS, Javascript)

→ Python, Java, C++

→ made web apps, android apps

→ worked with arduino & Raspberry Pi

↓

Helped with projects (3rd year, final year projects)

→ Automatic gate via RFID & Liscense plate detection

→ Automatic fish feeder

↳ dispenses certain mass of food at certain time (via formula)

↳ involves using load cells & motors.

↓

regards to PLC → Ladder diagram → confident that I can learn about PLC fast.

↓

Internship in Sejingkat Power Corporation (Coal fired power plant)

Exposed to Operator Interface System (OIS)

Learned about industry

→ SEB Power, State Dispatch Center (SDC)

→ Generation, Transmission, Distribution, Retail

→ Power plant, Generation Process, technical

Weekly equipment checkup

↳ operation

↳ standby

↳ maintenance

maintenance duties.

↳ learned a lot about the company, technical details, industry time management, communication skills.

↓

Substitute teacher → taught geo & Ask to f1 & 2 students

→ involved in marking papers & keying in results.

↳ increases communication skills (students, colleagues, superiors) time management, lesson planning.